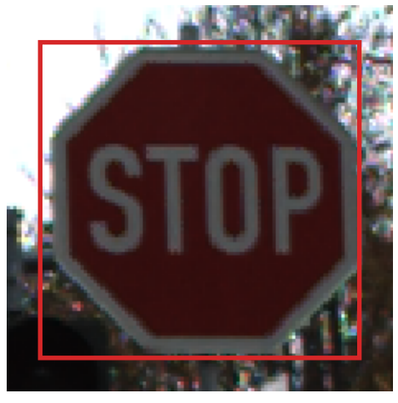
One GTSRB test image through the full pipeline (true class: Stop, test image 01020.ppm)

1 Original photo (111x111)



2 Crop to sign box (93x92)



3 Resize to 32x32



4 Median filter 3x3
(removes isolated specks)



5 CLAHE on lightness
(evens out brightness)



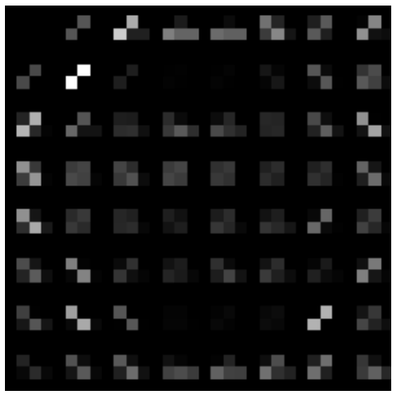
6 Grey image



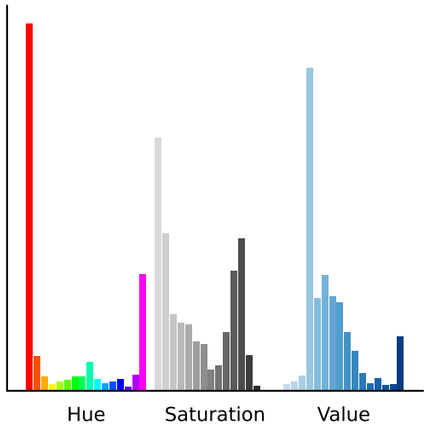
7 Edge strength
(brightness gradient)



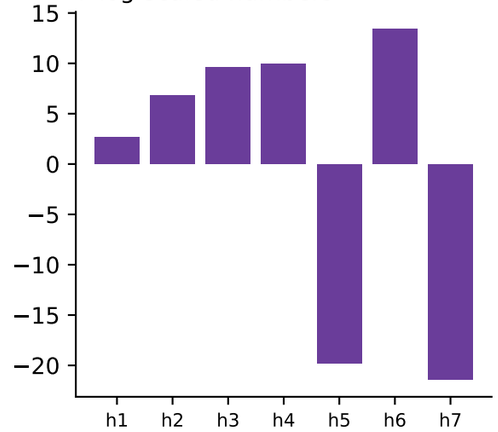
8 HOG: dominant edge
direction in each 4x4 cell



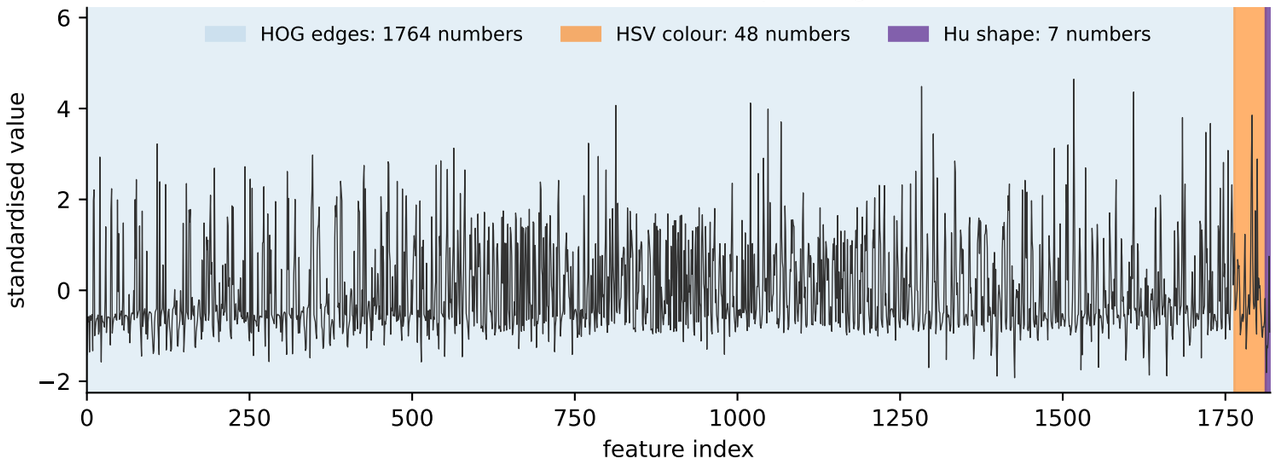
9 HSV colour histograms
(16 bins each)



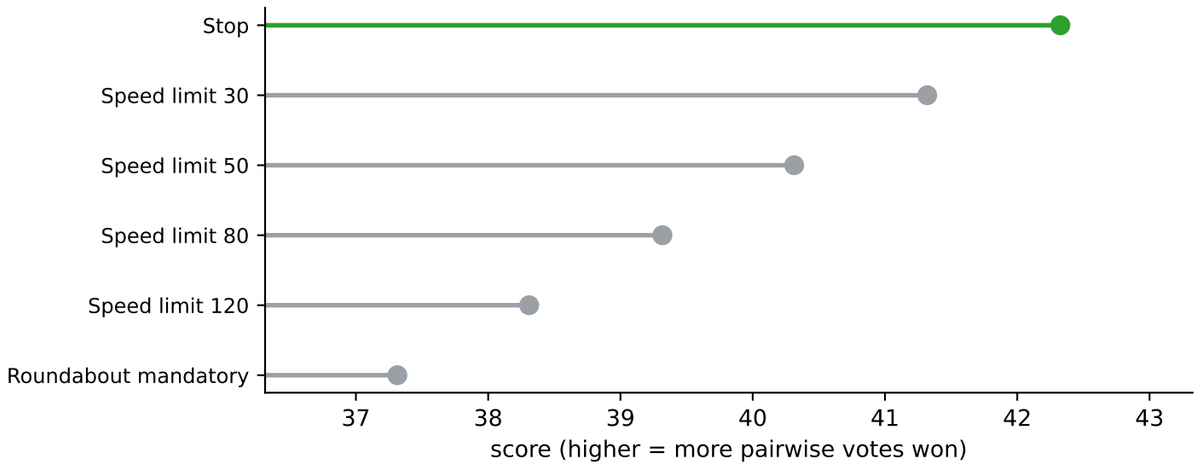
10 Hu moments (shape)
7 log-scaled numbers



11 Feature vector after standardisation: 1819 numbers per image



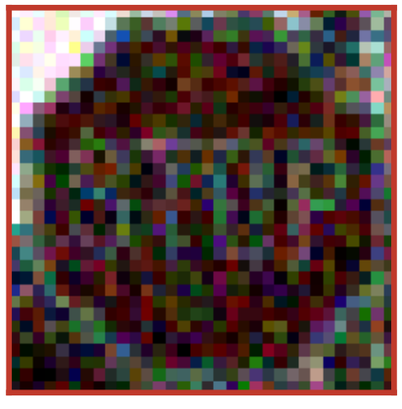
12 SVM class scores (pairwise votes), top 6
Predicted: Stop True: Stop



Clean image
Predicted: Stop
CORRECT



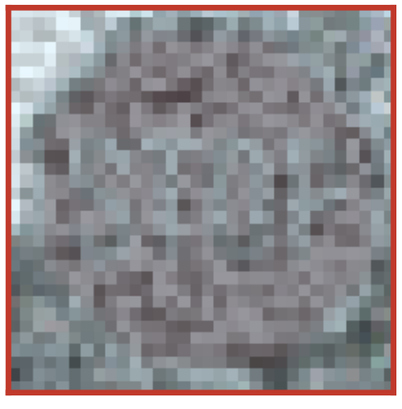
Gaussian Noise, severity 3
Predicted: Roundabout mandatory
WRONG



Motion Blur, severity 3
Predicted: Stop
CORRECT



Frost, severity 5
Predicted: Road work
WRONG



13 Robustness test

The same image is damaged, then steps 4 to 12 are repeated on the damaged picture.

The edge and colour numbers change, so the SVM's choice can change.